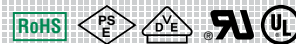


462 Series Fuse



Agency Approvals

AGENCY	AGENCY FILE NUMBERS	AMPERE RANGE
	E67006	500mA - 4A
	40022235 40027839	1A, 1.6A, 2A, 3.15A, 4A
	JET1896-31007-1005	2.5A - 5A
	E242325	1A - 5A

Electrical Characteristics for Series

% of Amp Rating	Opening Time
125%	1 hour, Minimum
200%	2 minutes, Maximum
1000%	10 milliseconds, Minimum 100 milliseconds, Maximum

Description

The 462 series Nano²® Surface Mount Fuse has time-lag current characteristics with interrupting ratings rated at 250V and 350V. It complies with IEC 60127-4 Universal Modular Fuse-Links.

Features

- Heat resistant plastic housing, UL 94 V-0
- Designed for line or low voltage applications
- Low voltage drop
- Internationally approved
- High pulse resistance
- Lead-free – compatible with lead-free solders and higher temperature profiles
- Available in ratings of 500mA to 5A

Applications

- Lighting ballast
- AC/DC adaptor primary protection
- Transformerless AC/DC converter circuit
- High DC voltage power distribution system

Additional Information



Datasheet



Resources



Samples

Electrical Specifications by Item

Ampere Rating (A)	Amp Code	Max Voltage Rating (V) ⁵	Interrupting Rating	Nominal Cold Resistance (Ohms) ¹	Nominal Melting I ² t (A ² sec)	Nom Voltage Drop (mV)	Nom Power Dissipation (mW)	Agency Approvals ³			
											
0.500	0500	250	100A @ 350VAC/VDC ⁴	0.2270	0.43	160	200	X		X	
0.630	0630			0.1570	0.80	160	200	X		X	
0.800	0800			0.1300	1.40	160	250	X		X	
1.00	1100			0.0867	2.70	140	250	X	X	X	
1.25	1125			0.0602	5.20	130	250	X		X	
1.60	1160			0.0443	9.70	130	280	X	X	X	
2.00	1200			0.0335	5.44	120	300	X	X	X	
2.50	1250			0.0278	8.00	120	450	X		X	X
3.15	1315			0.0204	14.00	110	600	X	X	X	X
4.00	1400			0.0158	21.00	110	800	X	X	X	X
5.00	1500		150A @ 250VAC/VDC	0.0124	40.00	110	1000	X		X	X

1. Cold resistance measured at less than 10% of rated current at 23°C

2. I²t values slated for 8ms opening time

3. Agency Approval Table Key: X = Approved or Certified, P = Pending

4. UL Recognition - IR at 100A @ 350 VAC/VDC

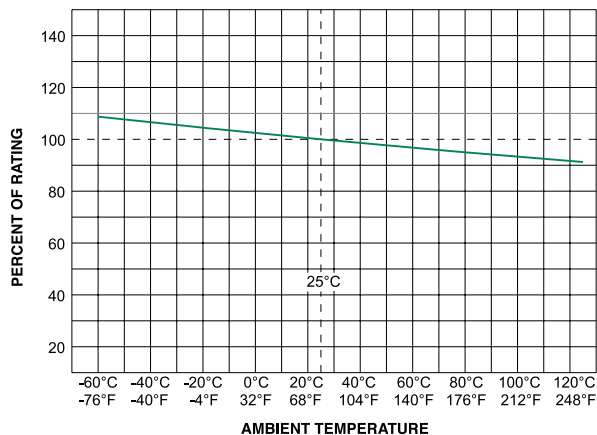
5. Rated at 350VAC/VDC per UL Recognition under UL248 (up to 4A only).
Rated at 250VAC/VDC per VDE under IEC standard 60127-4.

If you have special electrical characteristic needs, please contact Littelfuse to discuss application specific options.

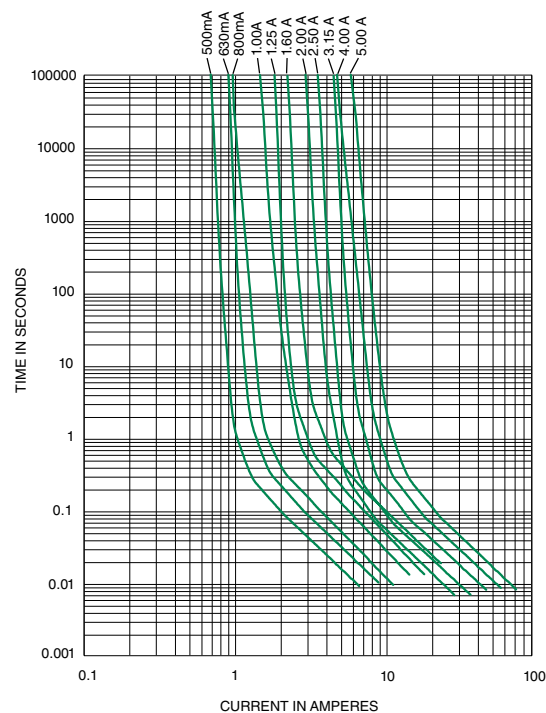
Surface Mount Fuses

NANO²® Fuse > 250V/350V VAC/VDC Time Lag > 462 Series

Temperature Derating Curve

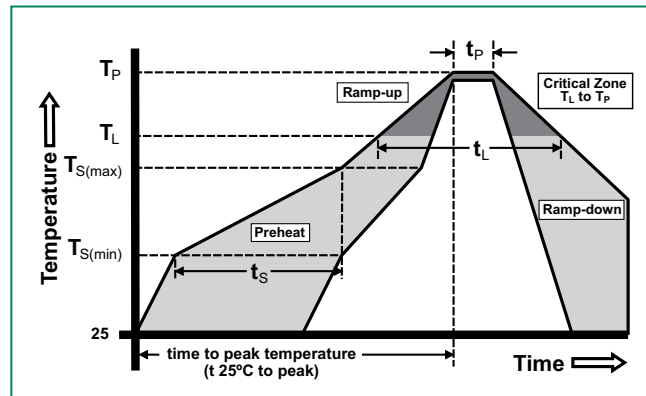


Average Time Current Curves



Soldering Parameters

Reflow Condition		Pb – free assembly
Pre Heat	- Temperature Min ($T_{s(min)}$)	150°C
	- Temperature Max ($T_{s(max)}$)	200°C
	- Time (Min to Max) (t_s)	60 – 120 seconds
Average Ramp-up Rate (Liquidus Temp (T_L) to peak)		5°C/second max.
$T_{s(max)}$ to T_L - Ramp-up Rate		5°C/second max.
Reflow	- Temperature (T_L) (Liquidus)	217°C
	- Time (t_L)	60 – 90 seconds
Peak Temperature (T_p)		250 ^{+0/-5} °C
Time within 5°C of actual peak Temperature (t_p)		20 – 40 seconds
Ramp-down Rate		5°C/second max.
Time 25°C to peak Temperature (T_p)		8 minutes max.

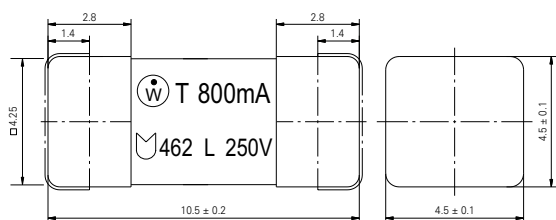


Product Characteristics

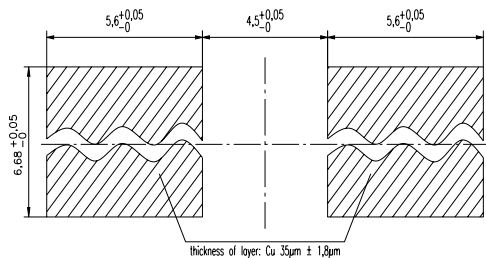
Materials	Body: Plastic UL 94 V-0 Cap: Tin-plated brass
Product Marking	Body: Brand Logo, "T" for Time-Lag, Current Rating, L Voltage Rating, UMF logo
Solderability	IEC 60068-2-58
Resistance to Soldering Heat	IEC 60068-2-58

Operating Temperature	-40°C to +85°C with proper derating
Climatic Category	IEC60068-1, -2-1, -2-2, -2-78 (-40°C to +85°C / 21 days)
Vibration	IEC60068-6 (24 cycles of 15 mins each, 1-60 Hz at 0.75mm amplitude, 60-2000 Hz at 10g acceleration)
Moisture Sensitivity Level	Level 1 J-STD-020C

Dimensions



Recommended Pad Layout



Part Numbering System

462	0	500	0	0	0	0
SERIES						RESERVE CHARACTER
AMP CODE						0: Default
0: < 1A						PACKAGING CODE
1: > 1A						0: Tape and Reel, 1500 pcs
AMP RATING						VARIANT
100: 1A						0: Standard
125: 1.25A						KIND
160: 1.6A						0: Standard
200: 2A						
250: 2.5A						
315: 3.15A						
400: 4A						
500: 500mA, 5A						
630: 630mA						
800: 800mA						

Examples:

0.5 amp (500mA) product is

462 **0 500** 0 0 0 0

5.0 amp product is

462 **1 500** 0 0 0 0

Please refer to Amp Code column of the Electrical Specifications table on the first page of this document.

Packaging

Packaging Option	Packaging Specification	Quantity	Quantity & Packaging Code
16mm Tape and Reel	IEC 60286, part 3	1500	0